__/__/____

In digital Image processing, an Image is defined as a two-dimensional function

$$f(x,y)$$

Where

(x,y) are the spatial coordinates (position of pixel)

$f(x,y)$ is the intensity value (gray level or color) at that point

(Amplitude of pixel)

Image is divide into two types.

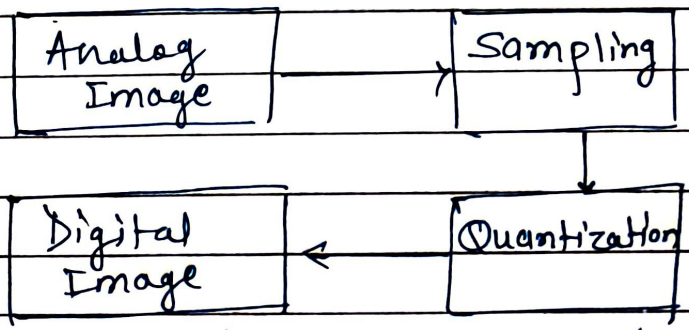
1. Analog Image
2. Digital Image

Analog Image: If all the value of x, y of f are continuous

Digital Image: If all the value of x, y of f are discrete and finite

Conversion of Analog Image to Digital Image

Why: We convert an analog image to digital Image so that it can be stored, processed and analyzed and transmitted using computer and digital systems



Conversion of Analog to Digital by using two fundamental steps Sampling and Quantization.

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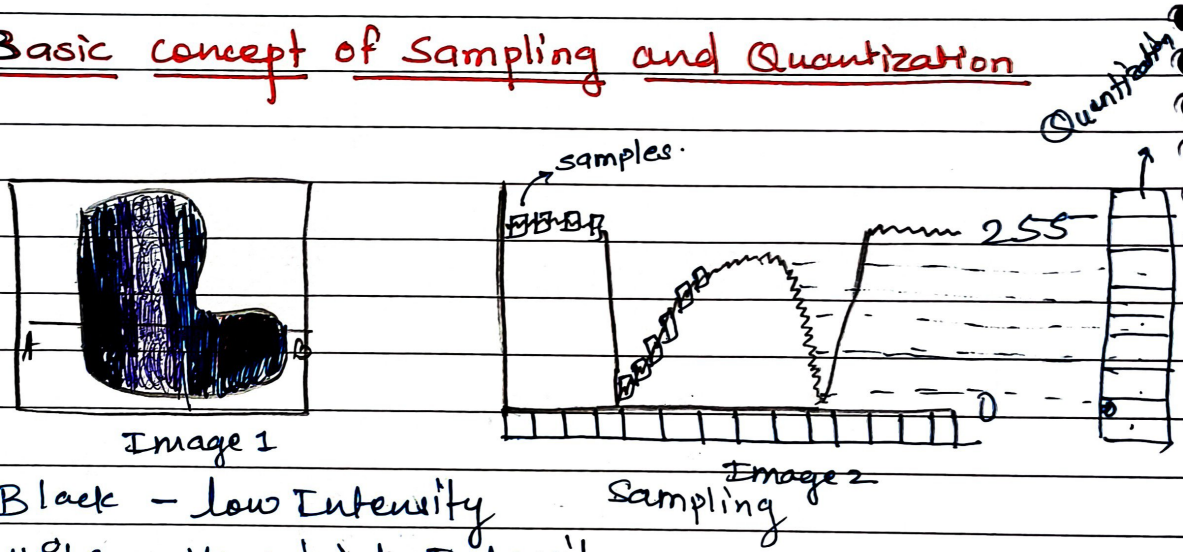
Sampling : Digitizing the coordinate value is called sampling

- It is a process of converting continuous spatial coordinates of an image into a discrete set of pixel

Quantization : Digitizing the amplitude value

- Quantization is the process of converting continuous intensity (amplitude)

Basic concept of sampling and Quantization



Black - low Intensity
White - Very high Intensity.

~~Digitization of this sample~~

Here Image 1 is a continuous Image.
Image 2 is one dimension Image.
Here samples are shows as square box
samples correspondence to this spatial location
samples are given digitised value on this vertical line

* Digitized samples based on the spatial location is called sampling

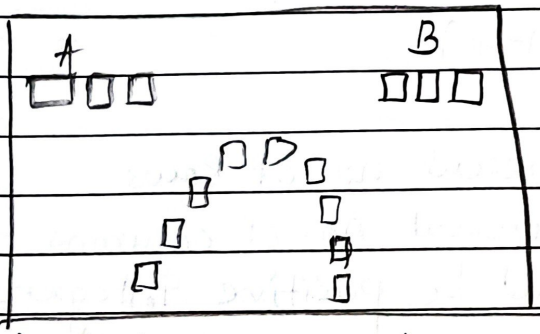
__/__/__

Now this samples can also be digitised with respect to intensity level

So there are 8 intensity level

Here each sample assigned with corresponding intensity value - which is called as quantization.

So the Resultant Image.



So in this way sampling and quantization is applied to a on analog image to obtained an digital Image

Representation of Digital Image :

Consider continuous Image

let $f(s,t)$ where

s and t are continuous variable.

2D array { M - Rows } which form digital image
N - column }

that digital Image is $f(x,y)$

Where f, x, y all are finite discrete quantity.

where $x \rightarrow$ varies from $0, 1, 2, \ldots m-1$

$y \rightarrow$ varies from $0, 1, 2, \ldots n-1$

x - Represent no. of rows

y - Represent no. of column

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- Basically there are three different way to represent digital Image.

- $f(x,y)$ can be represented by matrix

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \cdots & f(0,N-1) \\ f(0,1) & & & & \\ \vdots & & & & \\ f(m-1,0) & - & - & - & f(m-1,N-1) \end{bmatrix}$$

where M represent no. of rows
N represent no. of columns
M and N should be positive integers
but
always No. of intensity levels $L = 2^k$
where k is an +ve integer.

- * Dynamic Intensity Range : Is a ratio of

$$\frac{\text{maximum measurable intensity level}}{\text{minimum detectable Intensity level}}$$

maximum measurable intensity level ---→ saturation
minimum detectable Intensity level ---→ Noise.

which describe the max & min intensity Range

* So the intensity value beyond the saturation level is clipped off.
* the below the noise level is not true intensity value.

No. of bits of Image

No. of bits required to store a particular Image

$$bits = M \times N \times K.$$

$$\boxed{b = M \times N \times K}$$

If no. of rows & no. of columns are same

then. $M = N$

$$\boxed{b = N^2 K}$$

S

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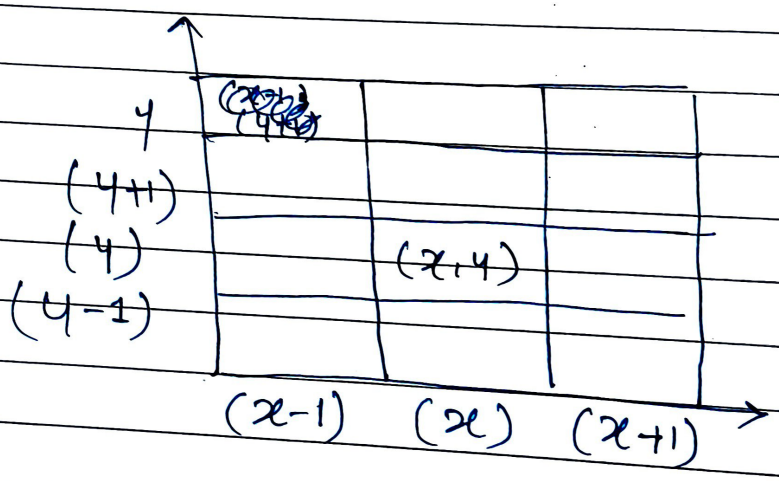
Relationship between Pixels

Neighborhood and Adjancy of Pixels.

1] Neighborhood of Pixels.

Any Image can be represented as the following



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Connectivity / Adjancy.

Two pixels that are neighbours and have the same gray level are adjacent.

a) 4 - adjacency. b) 8 - Adjacency.

Binary Image	Gray-Scale Image
0 1 0 1	54 10 100 8
0 0 1 0	81 150 2 34
0 0 1 0	201 200 3 45
1 0 0 0	7 70 147 56
V = {1}	V = {1, 2, 3, ---, 10}

c) M - adjacency.

```
0 1 1      0 1 1      0 1 1
0 1 0      0 1 0      0 1 0
0 0 1      0 0 1      0 0 1
```

```
0 1 1
0 1 0
0 0 1
```

Distance Measures between Pixels.

Let p and q be pixels with coordinates (x,y) and (s,t) respectively.

1) Euclidean Distance

$$De(p,q) = [(x-s)^2 + (y-t)^2]^{1/2}$$

OR.

$$De(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$

2) City Block Distance

$$D4(p,q) = |x-s| + |y-t|$$

Ex: The pixels with D4 distance ≤ 2 from (x,y) form the following contours of constant distance. (Shape)

```
        2
    2   1   2
2   1   0   1   2
    2   1   2
        2
```

```
    *
  * * *
    *
```

3) Chessboard distance

$$D8(p,q) = \max(|x-s|, |y-t|)$$

ex: The pixels with D8 distance ≤ 2 from (x,y) form the following contours of constant distance (shape)

2 2 2 2 2
2 1 1 1 2
2 1 0 1 2
2 1 1 1 2
2 2 2 2 2

4) Dm Distance :

It is defined as the shortest m-path between the points. This distance between 2 pixels depends on the values of the pixels along the path as well as the value of their neighbours.

P(x,y) q(s,t)

Note : The D4 and D8 distance between p and q are independent of any paths that might exist between the points because these distance involves only the coordinates of the points

Question :

The image segment is shown below. let V be the set of gray level values used to define connectivity in the image.
Compute D4, D8 and Dm distances between pixels 'p' and 'q' for.

i) V = {2,3}
ii) V = {2,6}

(0,0) (P) 2 3 2 6 1
6 2 3 6 2
5 3 2 3 5
2 4 3 5 2
4 5 2 3 6 (q) (4,4)

coordinates of P(x,y) = (0,0)
coordinates of q(s,t) = (4,4)

$D_4(p,q) = |x-s| + |y-t|$
$= |0-4| + |0-4|$
$= 4+4 = 8$ unit

$D_8(p,q) = \max(|x-s|, |y-t|)$
$= \max(|0-4|, |0-4|)$
$= \max(4,4)$
$= 4$ unit

(i) $V = \{2,3\}$. There is no path between p and q as $\notin q(6)$ is not included in the set V

(ii) $V = \{2,6\}$: There is no path between P and q

ex: 2 conside the following image segment. Compute Dm D_4 and D_8 distances between pixels 'p' and 'q' for $V = \{0,1\}$ where v is the set of gray level values used to define connectivity. Repeat for $V = \{1,2\}$

3 1 2 1 (q) (3,3)
0 2 0 2
1 2 1 1
(P) 1 0 1 2
origin (0,0)

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coordinates of $P(x,y) = (0,0)$

$q(s,t) = (3,3)$

$D_4(P,Q) = |x-s| + |y-t|$

$= |0-3| + |0-3|$

$= 3+3$

$= 6$ units.

$D_8(P,Q) = \max(|x-s| + |y-t|)$

$= \max(|0-3| + |0-3|)$

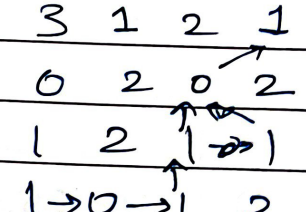
$= \max(3,3)$

$= 3$ units.

Dm

⇒ i) $V = \{0,1\}$ =

$D_m = 5$ units.



Dm

⇒ ii) $V = \{1,2\}$

$D_m = 6$ units.

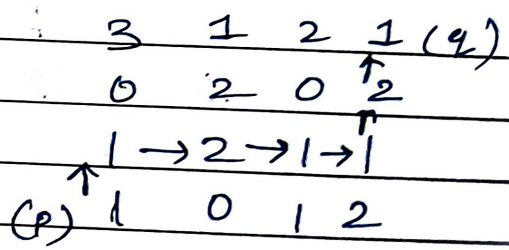


Image Enhancement

The process of manipulating the image so that the result is more suitable than the original for specific applications.

Techniques for Image Enhancement

(I) spatial Enhancement domain : Involves direct manipulation of pixels.

(II) Frequency Domain : Involves modifying the fourier transform of an Image.

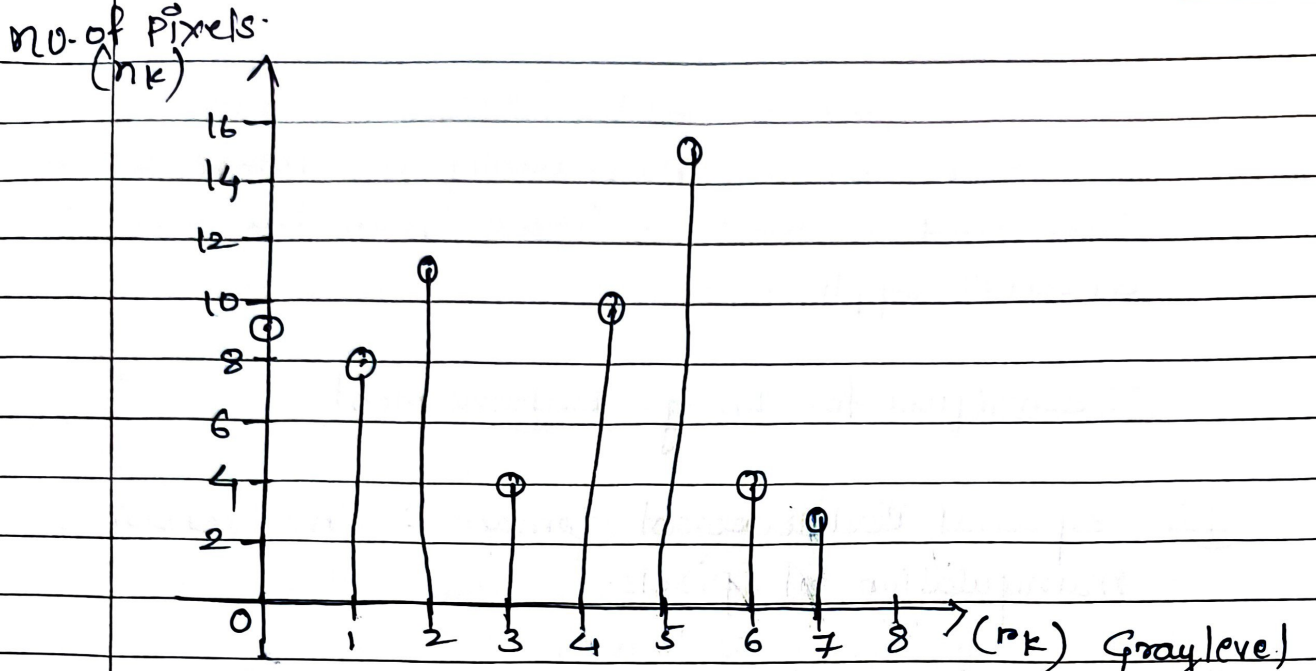
(III) combination : Involves a combination of spatial & frequency domain.

(I) Spatial Domain

(1)(1) Histogram Equalization : Histogram equalization in image processing is a contrast enhancement technique that redistributes pixels intensities to creat a more uniform histogram

Que1: Perform the histogram equalization for an 8X8 image shown below.

Gray levels	0	1	2	3	4	5	6	7
No. of Pixels	9	8	11	4	10	15	4	3.



Histogram of Input Image

Histogram equi. level	Gray level (r_k)	No. of Pixels (n_k)	$P(r_k) = \frac{n_k}{n}$ (PDF) (Probability Distribution fun.)	Sk (CDF)	Sk x 7
≈ 1	0	9	9/64 = 0.141	0.141	0.987
2	1	8	8/64 = 0.125	0.266	1.862
3	2	11	11/64 = 0.172	0.438	3.066
4	3	4	4/64 = 0.0625	0.5005	3.5035
5	4	10	10/64 = 0.156	0.6565	4.5955
6	5	15	15/64 = 0.234	0.8905	6.2335
7	6	4	4/64 = 0.0625	0.953	6.671
7	7	3	3/64 = 0.047	1	7
		n=64			

Cumulative distribution function.

Gray level	1	2	3	4	5	6	7
No. of Pixels	9	8	11	4	10	15	7

L = Total no. of gray levels in the Image
for k-bit Image $L = 2^k$.

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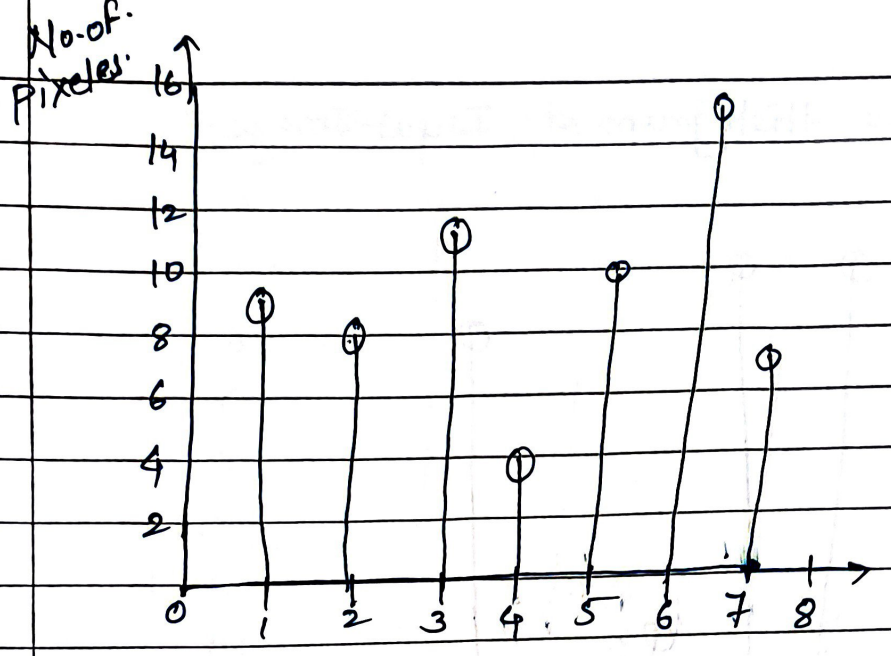


Histogram of processed Image

Q2: Peform Histogram equalization for the following Image

$f(x,y) =$

1	2	1	1	1
2	5	3	5	2
2	5	5	5	2
2	5	3	5	2
1	1	1	2	1

max value = 5

Input Image.

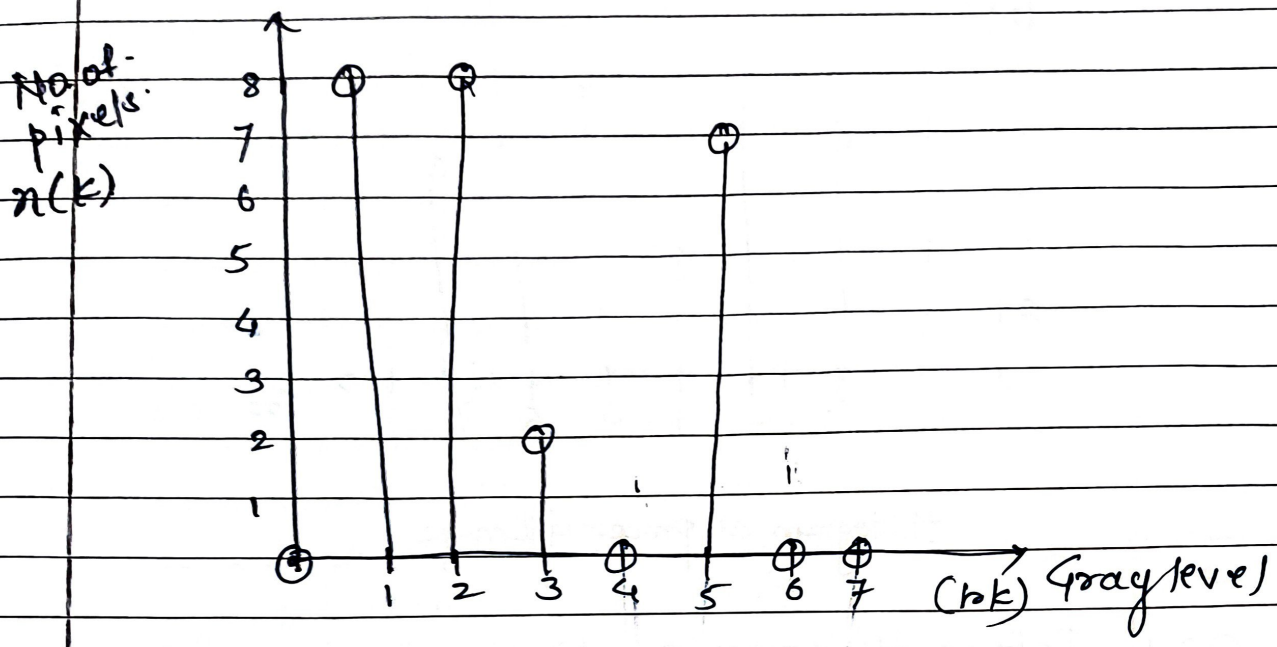
Sol^n: Now we need to calculate Interval. [0 L-1] (L = intensity level)

$2^0 = 1$ $2^1 = 2$ $2^2 = 4$ $2^3 = 8$ $L = 8$ $L-1 = 7$

Gray level.	0	1	2	3	4	5	6	7
No. of pixels	0	8	8	2	0	7	0	0

__/__/____

How draw Histogram of Input Image.

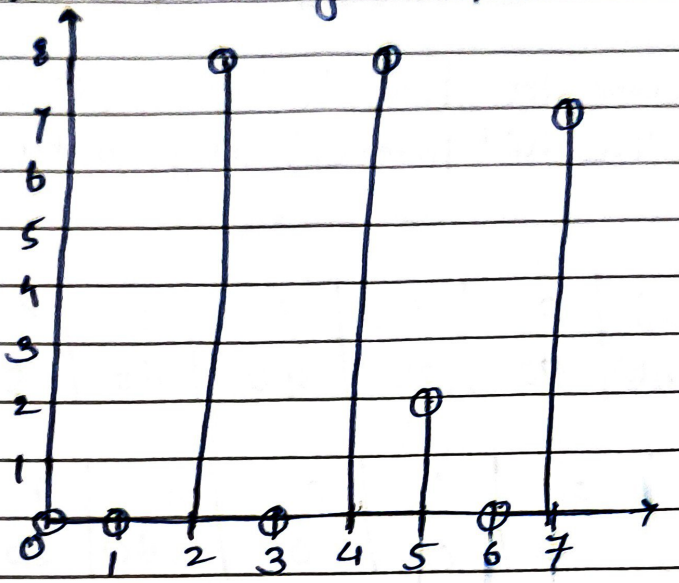


Gray level (rk)	No. of Pixels	$P(rk) = nk/n$ PDF	Sk CDF	Sk×7	Histogram equilization level
0	0	0	0	0	0
1	8	0.32	0.32	2.24	2
2	8	0.32	0.64	4.48	4
3	2	0.08	0.72	5.04	5
4	0	0	0.72	5.04	5
5	7	0.28	1	7	7
6	0	0	1	7	7
7	0	0	1	7	7.

$n = 25$

Gray level	0	2	4	5	7
Pixel values	0	8	8	2	7

How the Histogram of Procased Image.



How to calculate the output Image

1	2	1	1	1
2	5	3	5	2
2	5	5	5	2
2	5	3	5	2
1	1	1	2	1

Input Image

→

2	4	2	2	2
4	7	5	7	4
4	7	7	7	4
4	7	5	7	4
2	2	2	4	2

Output Image.

__/__/__

(2) Histogram Matching: (Specification)

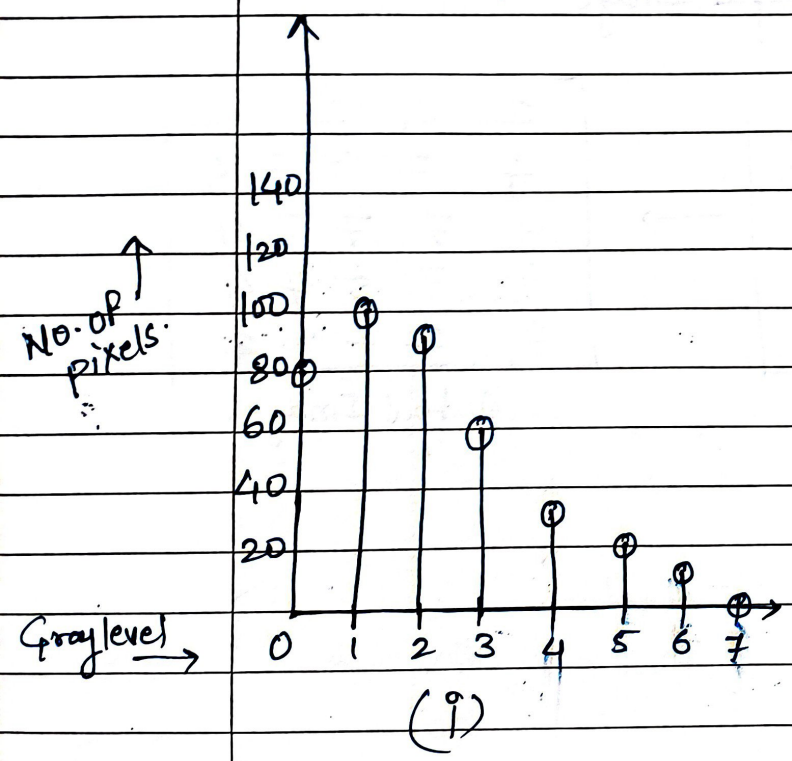
Q1. Given below are two histogram (i) and (ii) modify the histogram (i) as given by histogram (ii)

i)

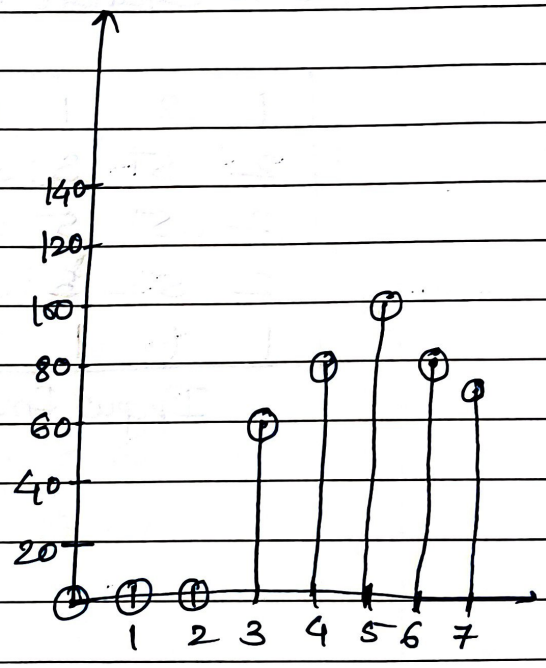
Gray level (r_k)	0	1	2	3	4	5	6	7
No. of Pixels (n_k)	80	100	90	60	30	20	10	0

ii)

Gray level (r_k)	0	1	2	3	4	5	6	7
No. of Pixels (n_k)	0	0	0	60	80	100	80	70



(i)



(ii)

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Step 1 Equalize Histogram (i)

Gray level	n_k	$P(r_k) = n_k/n$ (PDF)	s_k (CDF)	$s_k \times 7$	Histogram equalization	New. (n_k)
0	80	0.20	0.20	1.4	1	80
1	100	0.25	0.45	3.15	3	100
2	90	0.23	0.68	4.76	5	90
3	60	0.15	0.83	5.81	6	90
4	30	0.07	0.9	6.3	6	
5	20	0.05	0.95	6.65	7	30
6	10	0.02	0.97	6.79	7	
7	0	0	0.97	6.79	7	

$n = 390$

Step 2: Equalize Histogram (ii)

Gray level	n_k	$P(r_k) = n_k/n$ PDF	s_k (CDF)	$s_k \times 7$	Histogram equalization
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	60	0.15	0.15	1.05	1
4	80	0.20	0.35	2.45	2
5	100	0.25	0.6	4.2	4
6	80	0.20	0.8	5.6	6
7	70	0.17	0.97	6.79	7

$n = 390$

__/__/__

Step3: **Mapping**

- Take 1st and last column of Histogram (ii)
- Take the last 2 column of Histogram (i)

(ii)

Gray level	Histogram equilization level
0	0
1	0
2	0
3	1
4	2
5	4
6	6
7	7

(i)

Histogram equilization level	new (nk)
1	80
3	100
5	90
6	
6	90
7	
7	30
7	

Gray level	0	1	2	3	4	5	6	7
No. of pixels	0	0	0	80	80	100	90	30

(II) Frequency Domain

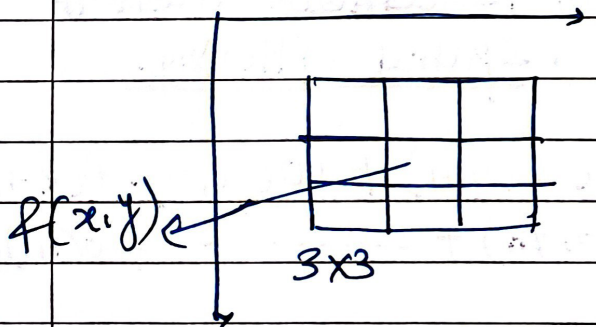
Image processing done by converting the image into frequency component and modifying them.

Spatial Filtering

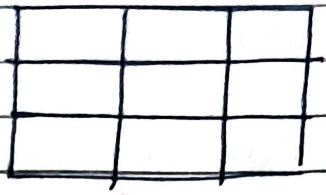
Histogram techniques improve the image globally. but sometimes we want to improve image locally pixel by pixel.
This needs spatial Filtering.

Spatial Filtering is an image enhancement technique in which pixel values are modified using neighboring pixels.

To perform operation on spatial filtering we use mask/ window/ kernel/ filter/ Template
it can be 3X3, 5X5, 7X7.
the elements of the mask is called coefficient



In spatial filter in order to perform operation on pixel neighbour we need to consider masks.



3X3 mask.

w(-1,-1)	w(-1,0)	w(-1,1)
w(0,-1)	w(0,0)	w(0,1)
w(1,-1)	w(1,0)	w(1,1)

mxn

-1 (Previous row, column) } mask coefficient
+1 (next row, column) }

f(x-1,y-1)	f(x-1,y)	f(x-1,y+1)
f(x,y-1)	f(x,y)	f(x,y+1)
f(x+1,y-1)	f(x+1,y)	f(x+1,y+1)

MXN

To perform the spatial filter operation we need to coonside the mask coefficient with image pixel so afterperforming this the resultant intensity value will be assign to the middle pixel.

If operation perform is linear then it is called as linear spatial filtering.

$$R = w(-1,-1).f(x-1,y-1) + w(-1,0).f(x-1,y) + w(-1,1).f(x-1,y+1) + \cdots\cdots + w(1,1).f(x+1,y+1)$$

In general equation :

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) + f(x+s, y+t)$$

Where mxn - size of mask $a = \frac{m-1}{2}$ $b = \frac{n-1}{2}$

MXN - size of image

and $x = 0, \ldots N-1$

No. of Rows in image

$y = 0 \ldots N-1$

No. of column in the image

In spatial filtering there are two terms Correlation and Convolution

Correlation: It is process of moving the mask over entire image and computing the sum of product of mask coefficient of image pixel of neighbour. and assign the coefficient value to the center of pixel

Example: 1D function.

f: 0 0 0 1 0 0 0 0

w: 1 2 3 2 8 $= m = 5$

$m - 1 = 4$

padding zeros.

0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0

1 2 3 2 8

Full correlation Result: 0 0 0 8 2 3 2 1 0 0 0 0

Crop correlation Result: 0 8 2 3 2 1 0 0

Convolution:

f: 0 0 0 1 0 0 0 0

w: 8 2 3 2 1.

0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0

8 2 3 2 1

Full convolution Result = 0 0 0 1 2 3 2 8 0 0 0 0

Crop convolution Result = 0 1 2 3 2 8 0 0

__/__/__

2D function : f(x,y)

```
0 0 0 0 0
0 0 0 0 0
0 0 1 0 0
0 0 0 0 0
0 0 0 0 0
```

filter (w)

```
1 2 3
4 5 6
7 8 9
```

⇒ Here we need to do zero padding

here in mask m = column = 3 - 1 = 2

n = rows = 3 - 1 = 2.

②

```
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 1 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
```

⇒

after correlation

```
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 1 2 3 0 0 0
0 0 0 4 5 6 0 0 0
0 0 0 7 8 9 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
```

after convolution.

```
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 9 8 7 0 0 0
0 0 0 6 5 4 0 0 0
0 0 0 3 2 1 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0
```

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General Expression of Correlation

$$w(s,t) \star f(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

General Expression of convolution

$$w(s,t) * f(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x-s, y-t)$$

$$a = \frac{m-1}{2} \qquad b = \frac{n-1}{2}$$

$m, n \rightarrow$ mask size

$M, N \rightarrow$ Image size.

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Correlation:

Correlation multiplies the mask with neighborhood without changing the mask.

ex: Image window (3X3) Msk window

1 2 3	1 0 -1
4 5 6	1 0 -1
7 8 9	1 0 -1

$(1\times1)+(2\times0)+(3\times-1)$
$+(4\times1)+(5\times0)+(6\times-1)$
$+(7\times1)+(8\times0)+(9\times-1)$
$=(1-3)+(4-6)+(7-9)=-6.$

Output pixl = -6.

Contrast stretching

Contrast stretching

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

- It stretches the contrast that is dynamic range of the image.
- Contrast stretching is essential when the image has a low contrast, i.e. the intensity values of the pixels in the image fall in a very narrow range.
- In the low contrast image specific details are difficult to determine due to the fact that most pixels are at the same intensity value.

⇒ Contrast stretching is the operation in which the pixel intensities are reassigned so that they can span a much broader range of intensities.

Contrast stretching is depends on slope of the particular range.

If slope=1, output image is same as input image
if slope>1, output image is brighter than input image
if slope<1, output image is darker (dull) than input image.

__/__/__

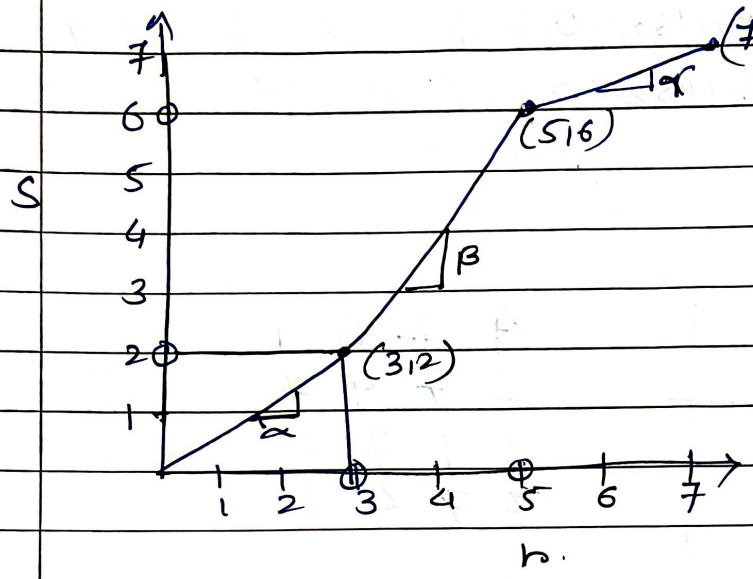
- Contrast stretching is a technique used to enhance the visual quality of an image by adjusting its intensity levels.
- The result is in a more visually appealing image, with enhanced contrast and improved visibility of details
- It corrects non-uniform lightning conditions and improves the visibility of features in an image.
- By applying contrast stretching, we can expand the dynamic range of the image and make it more visually appealing.

Contrast Stretching

4 3 5 2
3 6 4 6
2 2 6 5

Given $r_1 = 3$ and $r_2 = 5$

$s_1 = 2$ $\quad$ $s_2 = 6$



$$\alpha = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-0}{3-0} = \frac{2}{3} = 0.66$$

$$\beta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6-2}{5-3} = \frac{4}{2} = 2.$$

$$\gamma = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7-6}{7-5} = \frac{1}{2} = 0.5$$

$$S = \begin{cases} \alpha.r. & 0 \le r < 3 \quad r = 0, 1, 2, \\ \beta(r - r_1) + s1 & 3 \le r < 5 \quad r = 3, 4 \\ \gamma(r - r_2) + s2 & 5 \le r \le 7. \quad r = 5, 6, 7. \end{cases}$$

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r	s
0	$s = \alpha \cdot r = 0.66 \times 0$
1	$s = \alpha \cdot r = 0.66 \times 1 = 0.66 \approx 1$
2	$s = \alpha \cdot r = 0.66 \times 2 = 1.32 \approx 1$
3	$s = \beta(r - r_1) + s_1 = 2(3-3) + 2 = 2$
4	$s = \beta(r - r_1) + s_1 = 2(4-3) + 2 = 4$
5	$s = \beta_2(r - r_2) + s_2 = 0.5(5-5) + 6 = 6$
6	$s = \gamma_2(r - r_2) + s_2 = 0.5(6-5) + 6 = 6.5 \approx 7$
7	$s = \gamma_2(r - r_2) + s_2 = 0.5(7-5) + 6 = 7$
8	

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

4 2 6 1
2 7 4 7
1 1 7 6
7 7 4 1

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Spatial Domain Filters

Why do Image sometimes look noisy, blurred or unclear?

This naturally leads to Image enhancement.

Image enhancement is the process of improving the visual quality of an image for better interpretation
Image enhancement is used to

- Reduce noise
- Improve visibility of details.
- Highlights important features.

To enhance image, we process them in two main ways.

(1) Spatial Domain
- operates directly on pixels.
- pixels intensity values are modified.

$$g(x,y) = f(x,y) * h(x,y)$$

Where $f(x,y)$ is the original image
It represent the intensity value of the pixel at position (x,y)

Example: $0 \leq f(x,y) \leq 255$

$h(x,y)$ is the spatial filter.
also called as mask, kernel, template.

* convolution operator
- It place the mask over the image.
- Multiply corresponding values
- Add the result - Assign the value to the o/p pixel.

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$g(x,y)$ - is an output Image.

i.e. $f(x,y)$ = noisy image
$h(x,y)$ = mean filter
$g(x,y)$ = smoothed image

Smoothing and Sharping

⇒ Smoothing is the process of reducing noise and small intensity variation in an image by averaging neighboring pixels.

⇒ **Sharping:**
Sharping enhances edges and fine details in an image by emphasizing intensity difference.

How Smoothing in the Spatial Domain

- Pixel values are averaged with their neighbours.
- Sudden intensity changes are reduced.
- High-frequency components are suppressed.

It uses Spatial Domain Smoothing filters.

1. Mean filters.
2. Median filters.
3. Gaussian filter.

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Salt-and-Pepper Noise:

Salt and-pepper noise is a type of impulse noise where some pixels in an image are corrupted to

- Maximum intensity-(white) - salt.
- Minimum intensity-(black) - pepper.

Because the noise appears as random white and black dots scattered over the image, similar to sprinkled salt and pepper.

Example: 3x3.

Original 3x3

45	48	50
47	49	52
46	48	51

after ~~added~~ salt and peper noise.

45	255	50
47	49	0
46	48	51

255 - salt

0 - pepper.

Mean filter:

The mean filter replace the value of a pixel by the average of the pixel itself and its neighboring pixels.

Working of mean filter

Steps:

The mean filter is a spatial domain smoothing filter that replaces each pixel by the average of the pixels in its neighborhood.

Mathematically:

$$g(x,y) = \frac{1}{MN} \sum_{i=a}^{a} \sum_{j=b}^{b} f(x+i, y+j)$$

where

$M \times N$ = window size.

MN = total No. of pixels in the window.

if mean filter with 3×3 window

(1) window size

3×3

total pixel = 9

Mask.

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

(2) Window size 5×5

Total Pixel = 25

Mask:

$$\frac{1}{25} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

(3) Window size 7×7

Total pixel = 49

Mask =

$$\frac{1}{49}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Example on Next page:

Median Filter:

The median filter is a non-linear spatial domain filter that replaces the value of a pixel with the median (middle) value of the pixels in its neighborhood.

Working of median filter.

For each pixel:

1. Select a neighborhood window (usually 3×3 or 5×5
2. Arrange all pixels values in ascending order
3. Select the middle value (median)
4. Replace the center pixel with the median.

Ex:

45	48	50
47	255	49
46	48	47

step 1: Convert to 1D.

{45, 48, 50, 47, 255, 49, 46, 48, 47}

step 2: Sort the values

{45, 46, 47, 47, 48, 49, 50, 255}

total values are 9

Median position = 5th value

i.e. 48.

Now replace center pixel value 255 by 48.

→ Mean filter example

3X3 window

45	47	46
48	255	48
50	49	47

step1: Mean filter Mask (3X3)

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

step2: Add All pixel values

~~45+47+4~~

$(45\times1)+(47\times1)+(46\times1)+(48\times1)+(255\times1)+(48\times1)$

$= +(50\times1)+(49\times1)+(47\times1)$

$= 635$

step3: Divide by total No. of pixels

$\therefore$ New pixel value $= \frac{635}{9} = 70.55$

Roundoff $\doteq 71.$

$\therefore$ Center pixel is replaced by $\boxed{71}$

Gaussian Filter

A Gaussian filter is a linear spatial domain smoothing filter that reduces noise by weighted averaging where pixels closer to the center have highr weight.

- It is derived from the Gaussian (normal distribution) function,

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

where
σ = standard deviation

Gaussian filter Mask.

3X3 Gaussian mask $\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

Sum of coefficients = 1.

given window = $\begin{bmatrix} 45 & 47 & 46 \\ 48 & 50 & 48 \\ 50 & 49 & 47 \end{bmatrix}$

step1: Multiply with Gaussian mask.

$(45\times1)+(47\times2)+(46\times1)+(48\times2)+(50\times4)+(48\times2)+(50\times1)+(49\times2)+(47\times1)$

step2: Add these values

$45+94+46+96+200+96+50+98+47 = 772.$

step3: Divide by 16

New next pixel value = $\frac{772}{16} = 48.25$

final value = 48

__/__/__

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{\frac{-(x^2+y^2)}{2\sigma^2}}$$

x, y - distance from center pixel

σ - control the amount of smoothing

$e^{-(x^2+y^2)/2\sigma^2}$ - ensures weights decreases smoothly.

$\frac{1}{2\pi\sigma^2}$: Normalization.

Frequency Domain Filters

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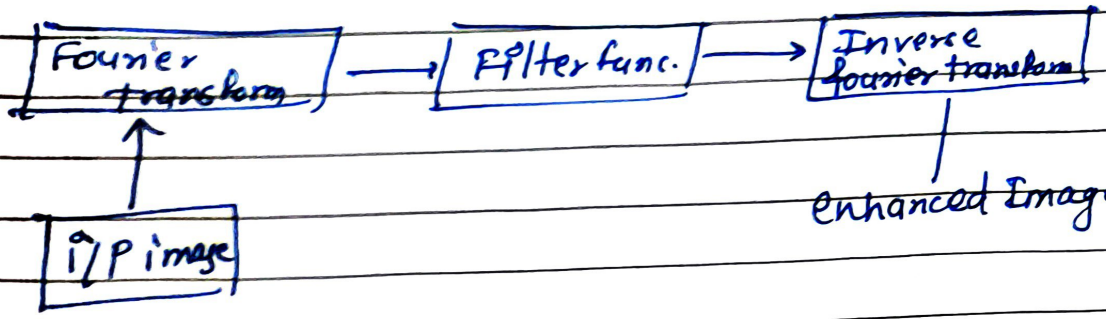


In spatial domain, we modify pixel values directly using mask (like mean filter etc)

In frequency domain
1. convert image into frequency components
2. Modify frequencies
3. convert back to image.

Mathematically:

$$g(x,y) = F^{-1}\{H(u,v) \cdot F(u,v)\}.$$

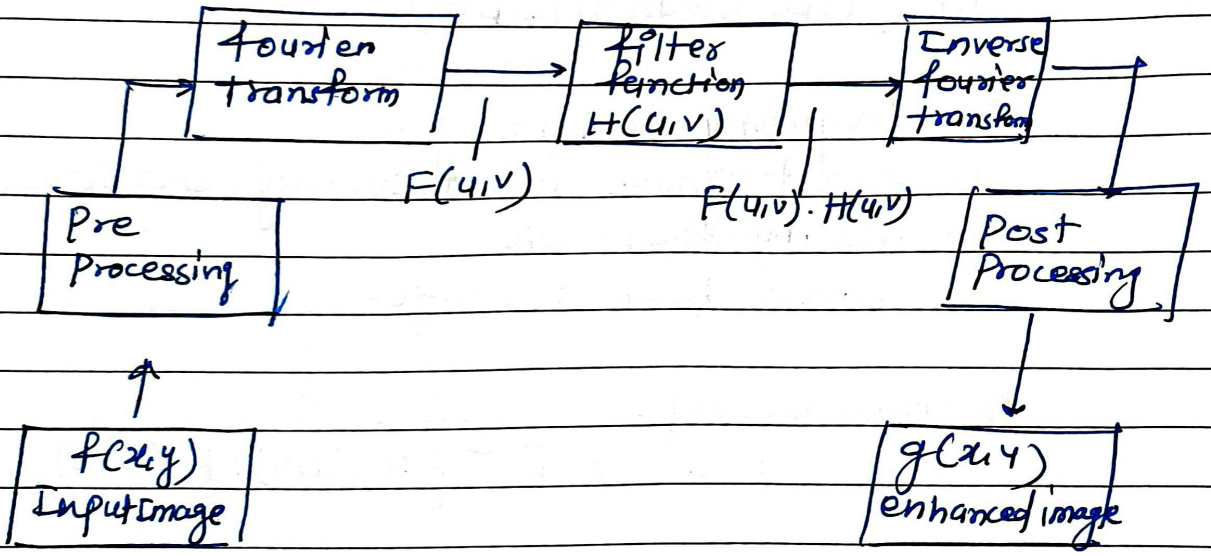
where:
$F(u,v)$ → fourier Transform of input image
$H(u,v)$ → Filter function
$g(x,y)$ → output image.

Frequency?
Low frequency → Smooth areas, background, slow intensity changes
High frequency → Edges, noise, sharp transitions.

__/__/__

Basic steps in Frequency Domain Filtering

1. Take image $f(x,y)$
2. Compute DFT $\rightarrow F(u,v)$
3. Multiply with filter $H(u,v)$
4. Take inverse DFT
5. Get enhanced image



Steps to solve the question

1) Multiply the input image by $(-1)^{x+y}$ to center the transform
2) Compute $F(u,v)$ the DFT of the image from (1)
3) Multiply $F(u,v)$ by Filter function $H(u,v)$
4) compute the inverse DFT of the result in (3)
5) obtain the real part of the result in (4)
6) Multiply the result in (5) by $(-1)^{x+y}$

$$\begin{bmatrix} (0,0) & (0,1) & (0,2) \\ (1,0) & (1,1) & (1,2) \\ (2,0) & (2,1) & (2,2) \end{bmatrix} \rightarrow \begin{bmatrix} (-1,-1) & (-1,0) & (-1,1) \\ (0,-1) & (0,0) & (0,1) \\ (1,-1) & (1,0) & (1,1) \end{bmatrix}$$

Smoothing frequency Filters

- These are also called low pass filters.
- These are used to smoother the image as they allow only low pass frequency components to pass through.
- These are used to remove noise.

Types of lowpass Filters

1. Ideal low pass filter (ILPF)
2. Butterworth lowpass filter
3. Gaussian lowpass filter

1. Ideal low pass filter (ILPF)

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

where D_0 is the cutoff frequency.

$D(u,v)$ - distance betn the point (u,v) from origin

2. Butterworth low pass filter (BLPF)

$$H(u,v) = \frac{1}{1 + [D(u,v)/D_0]^{2n}}$$

where D_0 is the cutoff frequency and n is the order of the butterworth filter.

3. Gaussian Low pass filter (GLPF)

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

where D_0 is the cutoff frequency

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Frequency Sharpening Filters

- These are also called high pass filters.
- These are used to sharpen the image as they only allow high frequency components to pass through.
- These are used to remove background of an image.

Types of high pass filters:

1. Ideal high pass filter (IHPF)
2. Butterworth high pass filter (BHPF)
3. Gaussian high pass filter (GHPF)

(1) Ideal High pass filter:

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

where D_0 is cutoff frequency

(2) Butterworth high pass filter

$$H(u,v) = \frac{1}{1 + \left[D_0 / D(u,v)\right]^{2n}}$$

where D_0 is the cutoff frequency and n is the order of the butterworth filter.

(3) Gaussian Highpass filter:

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

where D_0 is the cutoff frequency.

Here: $D(u,v) = \sqrt{\left(u - \frac{M}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2}$

Euclidean distance

ex 1 Convert the given spatial domain image using Fourier transform and perform Ideal lowpass filter to smoothen the image. Choose D_0 as 0.5.

1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0

Input Image

step 1: Multiply the Input Image by $(-1)^{x+y}$

	0	1	2	3
0	+1	-1	+1	-1
1	-1	+1	-1	+1
2	+1	-1	+1	-1
3	-1	+1	-1	+1

→ this is $(-1)^{x+y}$ matrix.

$W = e^{-j\frac{2\pi}{N}}$ $\quad e^{-j\pi} = \cos\pi - j\sin\pi$

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Now apply step 1 ⇒ multiply Input image by $(-1)^{x+y}$ matrix

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 \end{bmatrix}$$

step2: Compute the DFT of the image from (1)

$F(u,v) = \text{kernel} \times f(x,y) \times \text{kernel}^T$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hear we have converted image from ~~frequency~~ Spatial domain to frequency domain.

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step3: multiply $F(u,v)$ with the filter function $H(u,v)$

For Ideal low pass filter.

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

$$D(u,v) = \sqrt{\left(u - \frac{M}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2}$$

Here image is 4×4 $\therefore M=4 \quad N=4$

$$= \sqrt{\left(u - \frac{4}{2}\right)^2 + \left(v - \frac{4}{2}\right)^2}$$

$$= \sqrt{(u-2)^2 + (v-2)^2}$$

$(0,0)$ $(0,1)$ $(0,2)$ $(0,3)$
$(1,0)$ $(1,1)$ $(1,2)$ $(1,3)$
$(2,0)$ $(2,1)$ $(2,2)$ $(2,3)$
$(3,0)$ $(3,1)$ $(3,2)$ $(3,3)$

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Distance from each value of (u,v) from $(2,2)$

$(0,0) \rightarrow \sqrt{(0-2)^2 + (0-2)^2} = \sqrt{8} = 4.24$

$(0,1) \rightarrow \sqrt{}$

$(0,2)$

$(0,3)$

$(1,0)$

$(1,1)$

$(1,2)$

$(1,3)$

$(2,0)$

$(2,1)$

$(2,2)$

$(2,3)$

$(3,0)$

$(3,1)$

$(3,2)$

$(3,3)$

$$D(u,v) = \begin{bmatrix} 4.24 & 2.23 & 2 & 2.23 \\ 2.23 & 1.41 & 1 & 1.41 \\ 2 & 1 & 0 & 1 \\ 2.23 & 1.41 & 1 & 1.41 \end{bmatrix}$$

$D_0 = 0.5$ given.

$$\therefore H(u,v) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$G(u,v) = F(u,v) * H(u,v)$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 4. Compute the inverse DFT of the result in (3)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$ — IDFT kernel.

$$G^{-1}(u,v) = \frac{1}{4} \text{kernel} \times \text{Image} \times \frac{1}{4} \text{kernel}^T$$

$$= \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \times \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

$$= \frac{1}{16} \begin{bmatrix} 8 & 8 & 8 & 8 \\ -8 & -8 & -8 & -8 \\ 8 & 8 & 8 & 8 \\ -8 & -8 & -8 & -8 \end{bmatrix}$$

Step 5: Multiply the result by $(-1)^{x+y}$

$$g(x,y) = \frac{8}{16} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$ → final enhanced image

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Image segmentation

* Image segmentation is defined as a term in which we divide the image in various region.

* The dividing of an image in severed region is called as image segmentation.

There are three types of image segmentation detection

1. Point detection
2. Line detection
3. Edge detection

Point detection: When an discontinuity takes place in an image that just use to show a single point which is different from the all pixel values so, that type of detection we can say that is a point detection segmentation

e.g. 3×3 matrix with point detection

-1	-1	-1
-1	8	-1
-1	-1	-1

Here in image we can observe the point detection. Pixel in an image which is crossing the threshold value above the intensity and just a single pixel throughout the image.

So, we can say that the point in an image detected, hence it is called as point detection.

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(2) Line Detection :

When a discontinuity takes place through out the image which can be present in horizontal 45°, vertical or -45° of pattern. which is present in an line format the abrupt change in a pixel value of the image that making a line and make it different from the image is called as line detection in an image

line detection in an image can be present as follow.

-1	-1	-1
0	0	0
2	2	2

Horizontal

-1	-1	2
-1	2	1
2	1	1

45°

-1	0	2
-1	0	2
-1	0	2

Vertical

(3) Edge detection :

Edge detection method is mostly used to findi objeet boundries in an images.

It works by detecting discontinuities in brightness.

Edge detection is used for image segmentation and data extraction in area such as image processing, computer vision and machine vision.

common edge detection algorithms include sobel, canny, prewiet, roobents and fuzzy logic method.

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Image segmentation

Image segmentation is a computer vision technique used to divide an image into multiple segment or regions, making it easier to analyze and understand specific parts of the image. It helps identify objects, boundries and relevant features with in an image for further processing.

- This techniques is widely used in applications such as medical imaging, object detection, autonomous driving and image editing. By classifying each pixel into meaningful categories.

Types of Image segmentation

1. **Semantic segmentation**

Semantic segmentation involves classifying each pixel in an image into a specific class or category without distinguishing between different instances of the same class.

2. **Instance Segmentation.**

Instance Segmentation, on the other hand, aims to identify and distinguish individual objects or instances within an image.

3. **Panoptic Segmentation**: Panoptic segmentation is a hybrid method that combines both semantic and

Black - 0
White - 255

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Types of Edge detection operators

1. Gradient-Based Edge Operator.
2. Gaussian-Based Edge detection operator

1. Gradient-Based Edge Detection Operators

Gradient-based operators detect edges by measuring how quickly the pixel values or brightness changes in a image. These changes are called gradients.
If the change is large between two neighbouring pixels that indicate an edge.
They use first derivative to find edge.

2. Gaussian-Based Edge Detection Operator

Gaussian-based operators combine blurring and edge detection. They reduce the effect of noise by smoothing the image first and then look for changes using the second derivative.

There are more advanced and accurate than gradient-based operator.

Image Edge Detection Operators.

- Gradient Based
  - Sobel Operator
  - Prewitt operator
  - Roberto operator
- Gaussian Based
  - Canny edge detector
  - Laplacian of Gaussian

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

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1. **Sobel operator:**

The Sobel operator is a method used to detect edges in an image by checking how pixel values (brightness) change in the horizontal and vertical directions.

- It uses 3X3 kernels or masks which are convolved with the input image to calculate the vertical and horizontal derivative approximations respectively:

$$G_x = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix} \qquad G_y = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

Where

Mx - detect edges from left to Right

My - detect edges from top to bottom.

add both.

By combining the results from both it finds the overall edge strength and direction.

$$\begin{bmatrix} -2 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 2 \end{bmatrix}$$

$$|(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| +$$

$$|(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$

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2. **Prewitt Operator**

It is very similar to the sobel operator but with slight difference is that it calculates the edge gradients. Like sobel it detects edges in the horizontal and vertical directions using two 3X3 matrices but it uses a uniform averaging technique in its kernel making it less accurate than sobel but faster and simpler to implement.

$$G_x = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \qquad G_y = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_x = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \qquad G_y = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

* G_x detect edges from left to right.
* G_y detect edges from top to bottom.

add both

$$= \begin{bmatrix} -2 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

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3. Roberts Operator

It is one of the simplest edge detection methods as it focuses on how pixel value change diagonally in an image. The operator uses a 2x2 kernel to calculate the gradient of intensity between diagonally adjacent pixels and make it suitable for detecting edges that appear along diagonal directions.

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

G_x : detects diagonal edges that run from top-left to bottom-right.

G_y : detect diagonal edges that run from top-right to bottom-left.